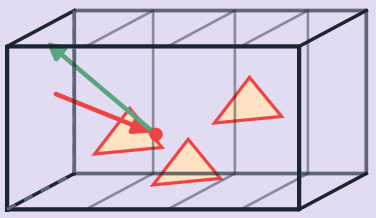
Ray Tracing Attention

Vocabulary geometry



Token / block identity

Output token

$s_{t+1} = \arg\max_v \ell_v$

Ray Tracing Readout

$\ell_v = \sum_b (P_b h)^T (P_b w_v)$

RMSNorm

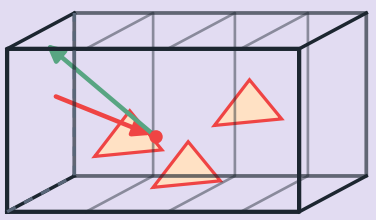
RTA layers

High-D layer state

$h = P^{-1}(\bigoplus_b P_b h)$

Restore / next-layer input

MLP coefficient field



Matrix tiles / static BVH

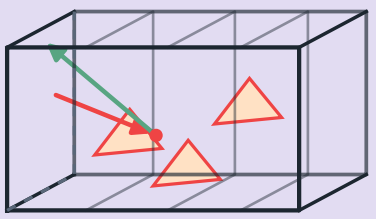
MLP

Matrix tiles / fixed surfaces
SwiGLU / full output

RMSNorm

$L \times$

Shared history BVH



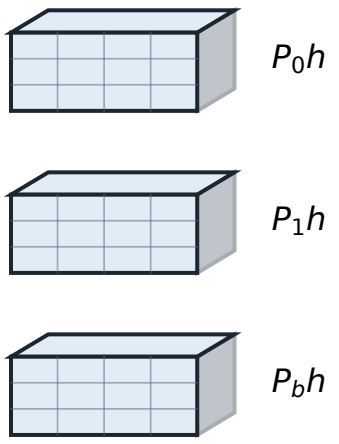
Q4_K keys / Q6_K values

Ray Tracing Attention

Projection / RoPE / shared query
Compact summaries / full readout

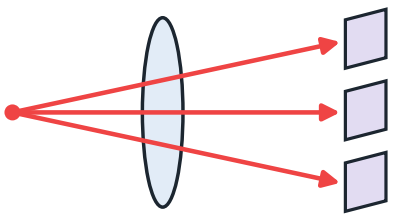
RMSNorm

Coordinate blocks



Complete high-D recovery

Parallel ray queries

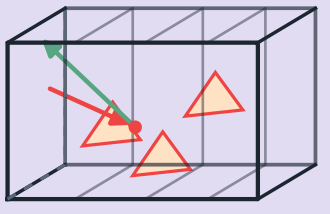


Compact full attention

$\bar{q}_a, \quad c_p$
$\mathcal{R}_B = (m_B, Z_B, \pi_B, U_B)$
$\pi_*, \quad o_d = U_{\pi_*(d)}/Z$

h_ℓ

Causal address BVH

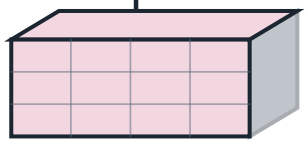


Token / successor / boundary

Query at each output step

Token embedding

$h_0 = E[s_{0:t}]$



$s_{0:t}$